1 FIG-218 — Median static maximum force by stage and checkpoint

Group: MD-FROM-SCRATCH-FAILURE-ATTRIBUTION **Evidence chain:** manifested **Status:** stage-wise MLIP summary; DFT validation pending

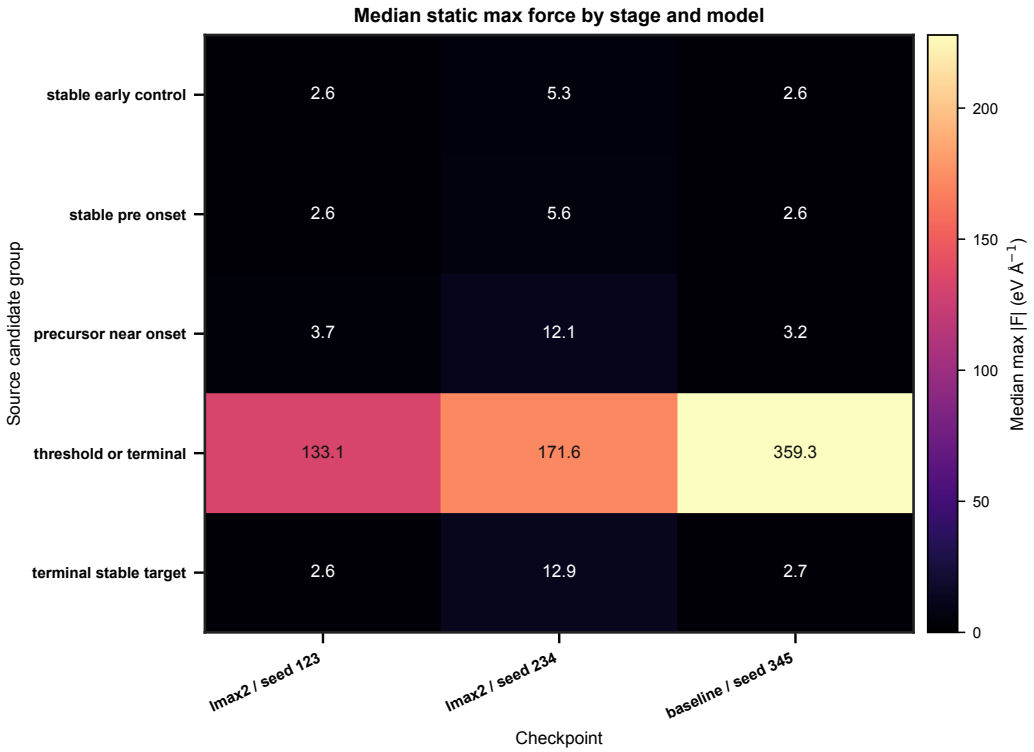


Figure 1: Median static maximum force for each source-stage group and registered checkpoint. The threshold-or-terminal row contains seven threshold candidates and is shown separately from the one terminal-stable target; source-stage labels are not recomputed from the static MLIP evaluations.

Results text draft. The stable-early and stable-pre medians remain near 2.6–5.6 eV Å^{-1}, while the precursor medians are 3.2, 3.7, and 12.1 eV Å^{-1} for baseline/seed 345, lmax-2/seed 123, and lmax-2/seed 234, respectively. Threshold-or-terminal medians increase to 133.1, 171.6, and 359.3 eV Å^{-1} in the same checkpoint order. **Discussion and boundary.** The heatmap summarizes a consistent separation between stable controls and threshold structures, with checkpoint-specific amplification in the precursor and threshold stages. It is evidence for a reproducible MLIP comparison on fixed structures, not evidence that one checkpoint is DFT-correct or that a deterministic physical onset time has been identified. **Article placement.** Proposed placement: Supplementary Information / stage-wise three-model MLIP force summary; no transport value is assigned to unstable branches. **Reviewer linkage:** overall status PENDING.

- **REV-C9 (PENDING)**: Inspect the concise stage-by-checkpoint summary. *Completed:* all stage groups and three model seeds are represented. *Still required:* reference-force validation and explicit reduced-cell parity checks.

Evidence pointers.

- Stage summary table: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/figures/table_mlip_model_stage_summary.csv`
- Analysis manifest: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/results/analysis_manifest.json`
- Plot script and Plot Atlas source: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/make_mlip_analysis.py`; `/Users/alina/Documents/Codex/plot_library`
- Asset SHA256: `8dfef46ae4c2419b50cd0e966416fa44c8eaa6b77233977d749a8aae942e939f`

Machine-collected evidence candidates (generated from the analysis manifest; do not edit this file).

- **Analysis manifest:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/results/analysis_manifest.json
- **Plot output manifest:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/figures/figure_and_text_manifest.json

Audit gap / next action: Await the selected DFT batch and assess whether reduced-cell forces preserve the qualitative MLIP stage ordering under convergence and PBC/parity checks.